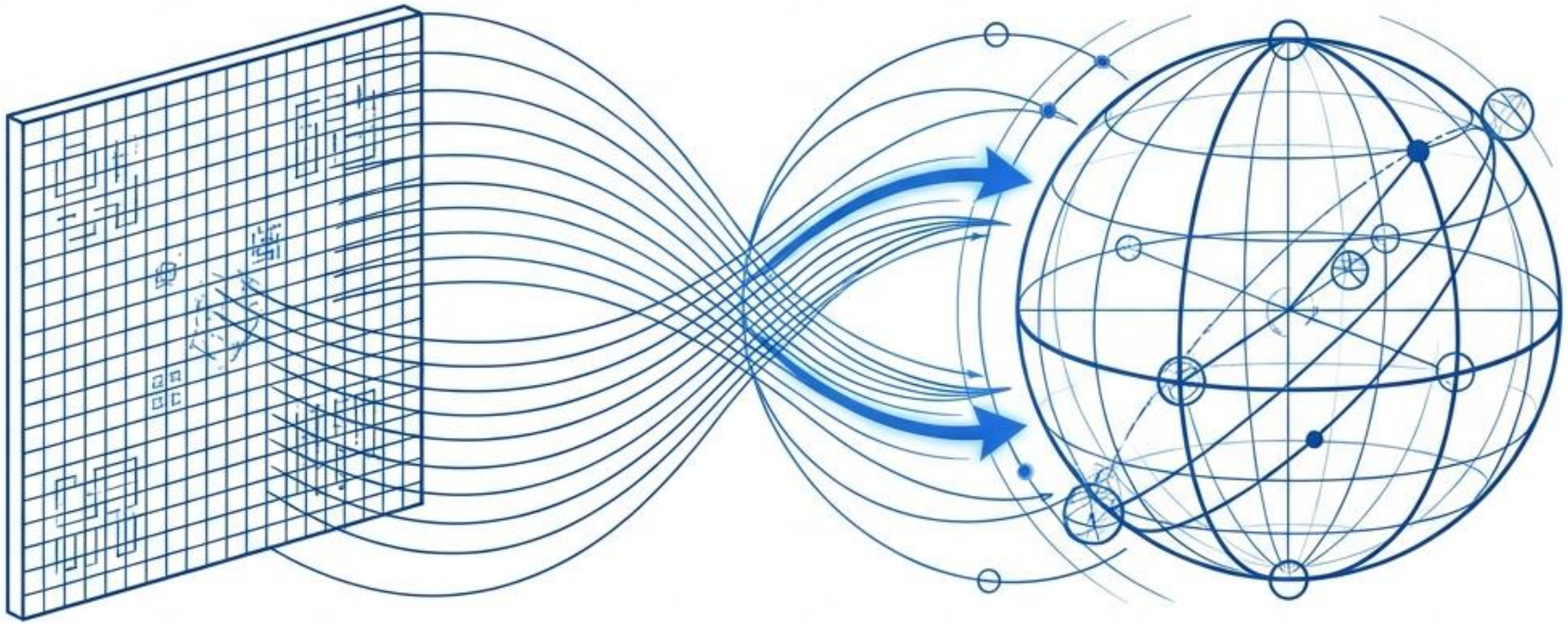
Multiclass Classification of Retina Datasets Using Variational Quantum Classifiers

A Hybrid Classical-Quantum Architecture for Diabetic Retinopathy Grading



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Aim of the Project

To build a **hybrid quantum-classical neural networks** that grade Diabetic Retinopathy on RetinaMNIST into **5 severity classes** using a **Variational Quantum Classifier**

- **Model A (baseline):** angle encoding, one feature per qubit rotation
- **Model B (upgrade):** amplitude embedding + strongly entangling layers, 64 features in 6 qubits
- Train both on a local simulator and benchmark them

Roadmap of This Deck

1. Background & tools
2. Quantum foundations
3. Classical CNN stage
4. Model A vs Model B
5. Training mathematics
6. Results & benchmarks

The Classical Bottleneck

Problem:
Diabetic Retinopathy (DR) grading on RetinaMNIST (1,600 highly imbalanced, 28x28 RGB images across 5 severity classes).

Limitation:
Classical Deep CNNs require massive datasets and suffer from over-parameterization and overfitting on subtle morphological lesions.

The Quantum Opportunity

Solution:
n qubits span a $2^{\wedge n}$-dimensional Hilbert space.

Advantage:
Variational Quantum Circuits (VQCs) learn complex, non-linear feature correlations with drastically fewer trainable parameters.

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The Qubit & Superposition

- A qubit state is a unit vector in $\mathbb{C}^2$
- **Superposition**: amplitudes on both $|0\rangle$ and $|1\rangle$
- Amplitudes are complex numbers
- **Born rule:** $|\text{amplitude}|^2$ = probability

State

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

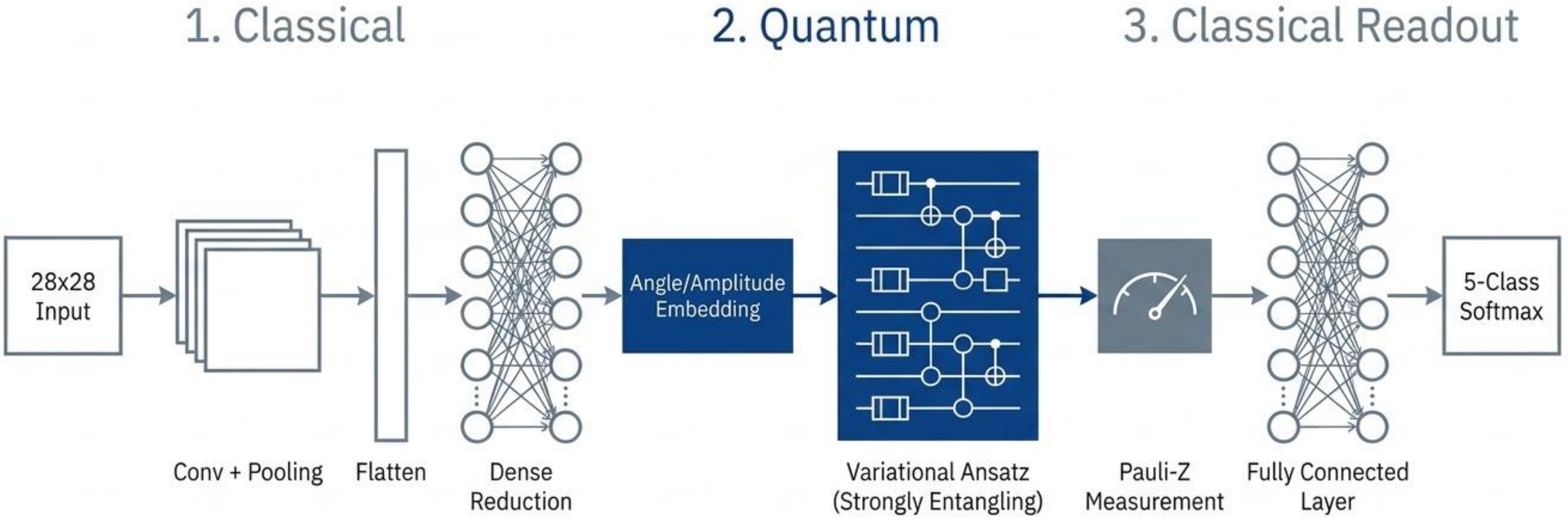
Normalization (Born rule)

$$|\alpha|^2 + |\beta|^2 = 1$$

Probabilities

$$P(0)=|\alpha|^2\,,\ P(1)=|\beta|^2$$

Memory trick: probabilities must add to 1, so the state vector has length 1.



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Stage 1: The CNN Backbone

- Image 28×28×3 = **2,352 values**: too many for NISQ qubits
- A CNN keeps only informative features
- Convolution slides a kernel over the image
- Output is compressed to n_features numbers

Convolution

$$y(i,j) = \Sigma\Sigma\, w(m,n)\cdot x(i+m, j+n) + b$$

```
self.conv1 = nn.Conv2d(3, 16, kernel_size=3)
self.flatten = nn.Flatten()
self.fc_reduce = nn.Linear(1568, n_features)
```

Fix applied: first conv takes 3 channels (RGB), not 1.

Memory trick: CNN = learned feature filter that shrinks 2,352 → n.

Stage 1: Dimensionality Funnel



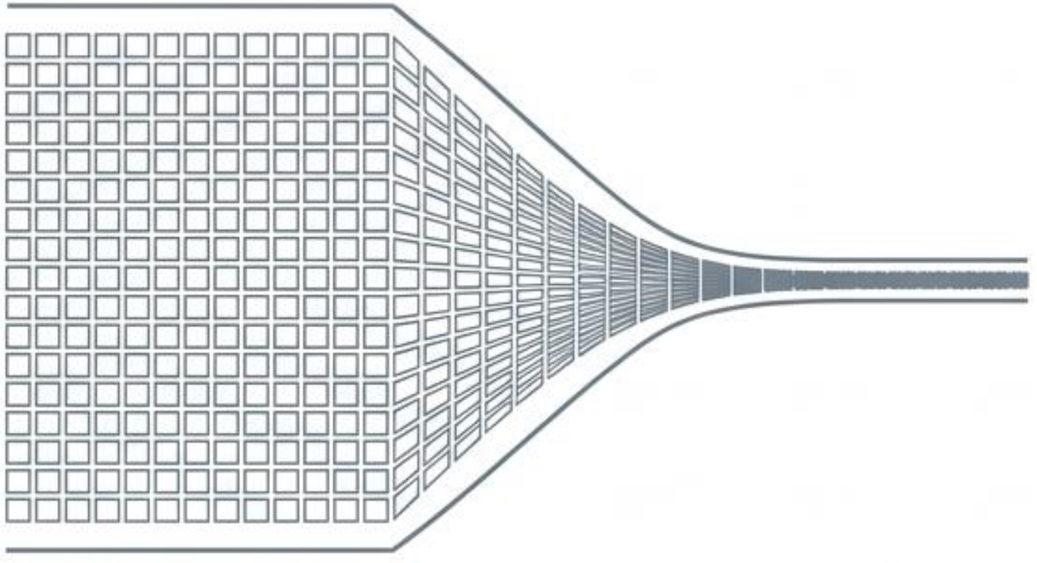
- Information is lost at the narrow end: **the bottleneck**
- Model A bottleneck = 6, Model B bottleneck = 64

Memory trick: the narrower the funnel, the more detail is thrown away.

Feature Extraction

Concept: 28x28x3 RGB images (2,352 values) cannot fit on raw NISQ devices. A CNN acts as a precursor bottleneck.

$$y_{i,j} = \sum \sum w_{m,n} x_{i+m,j+n} + b$$



Coding View: PyTorch CNN Backbone

```
self.conv1 = nn.Conv2d(3, 16, kernel_size=3)
self.flatten = nn.Flatten()
self.fc_reduce = nn.Linear(1568, n_features)
```

Fixed RGB channel mismatch

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Theory: Angle Embedding

Concept: One feature = One qubit rotation. Shallow depth, highly robust for noisy hardware.

$$|\psi(x)\rangle = \bigotimes_{i=1}^{n} R_x(x_i)\,|0\rangle$$

Matrix Breakdown:

$$R_x(\theta) = \begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$

Coding View

```
qml.AngleEmbedding(inputs,
wires=range(n_qubits),
rotation='X')
```

Limitation Note: Fails to capture sufficient medical detail for RetinaMNIST.

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The Bloch Sphere & Angular Encoding

Quantum State Representation: The Bloch Sphere & Angular Encoding

A single qubit is visualized as a point on a 3D unit sphere. By mapping continuous classical features directly into spherical rotation angles, classical data is translated into unique quantum states for quantum machine learning tasks.

3D Quantum State Geometry

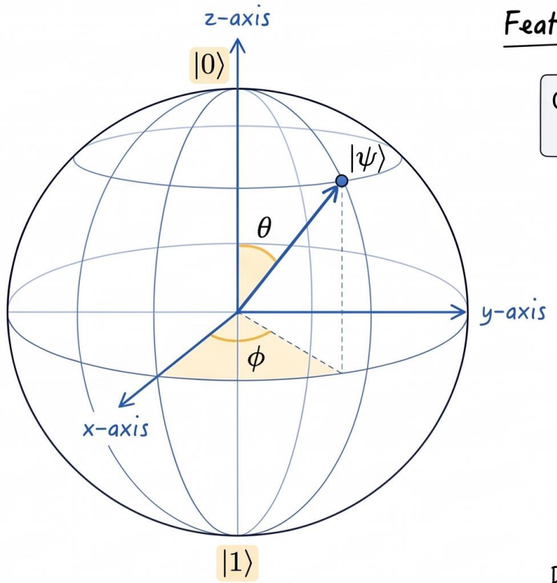
- A single qubit is represented geometrically as a point on a 3D unit sphere.

Latitude (θ) & Longitude (ϕ) Controls

- Latitude sets probability between $|0\rangle$ and $|1\rangle$.
- Longitude sets the relative quantum phase.

State Vector Formula

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$



Feature Mapping via Angular Encoding

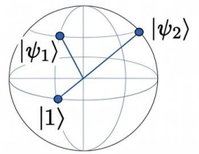
Classical Value	Rotation Angle
x	$\theta = f(x)$

Continuous Feature Substitution

Classical value x substitutes directly into rotation angle $\theta = f(x)$.

Unique Geometric Mapping

Every distinct real-valued number corresponds to a unique point on the sphere surface.



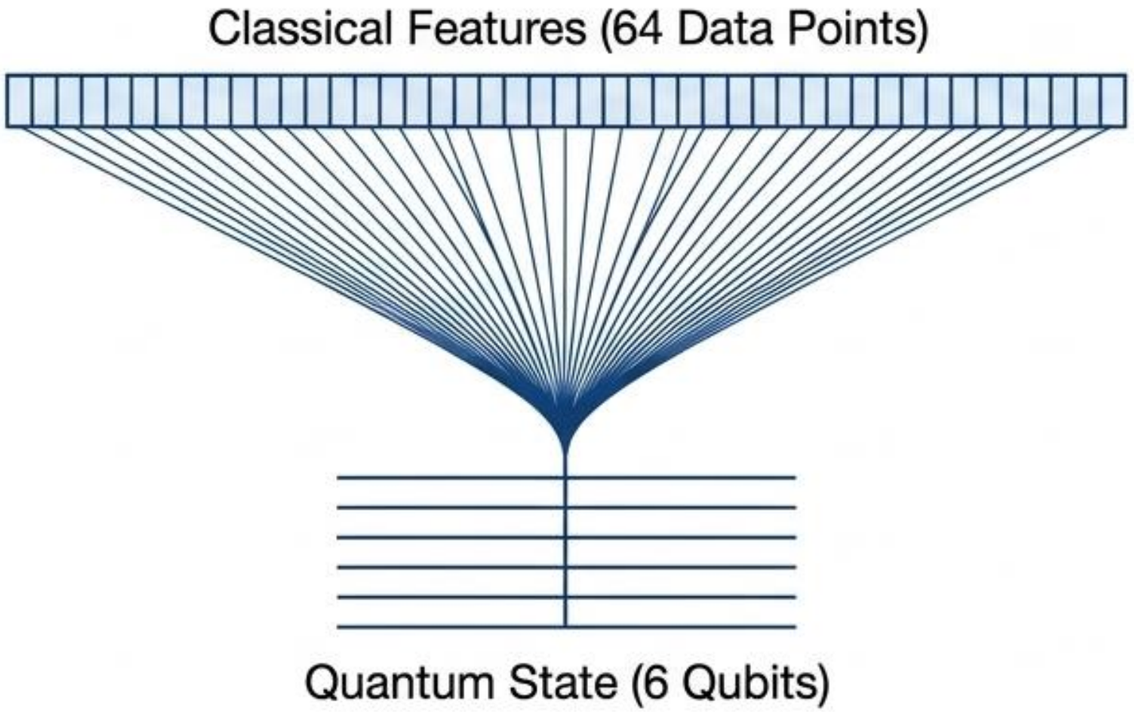
Distinct $x \rightarrow$ Unique $|\psi\rangle$

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- **θ** (latitude) sets probability of |0⟩ vs |1⟩
- **ϕ** (longitude) sets relative phase
- Feature x → angle θ = f(x)
- Distinct x → unique point on the sphere

Memory trick: North pole = |0⟩, south pole = |1⟩, a feature just tilts the arrow.

Amplitude Embedding



Benefit: Preserves critical micro-details (microaneurysms) without increasing NISQ hardware qubit demands.

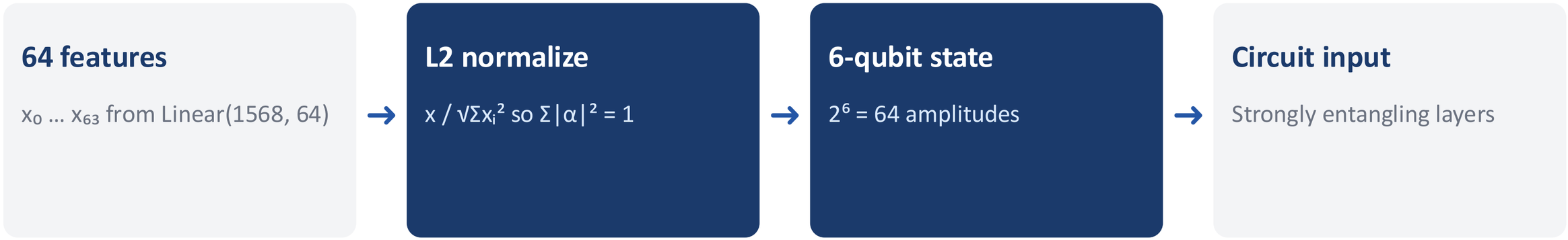
Coding View

```
# Bottleneck expanded from 6 to 64
self.fc_reduce = nn.Linear(1568, 64)


# PennyLane Interface
qml.AmplitudeEmbedding(features=inputs,
wires=range(n_qubits), normalize=True)
```

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Amplitude Embedding: Capacity Infographic



- **Benefit:** preserves critical micro-details without more NISQ qubits
- **Cost:** state preparation circuit is deeper than angle encoding

Memory trick: normalize first, embed second.

Model A: Angle Embedding

- **One feature = one qubit rotation**
- Shallow depth, robust on noisy hardware
- Capacity = number of qubits
- Only 6 numbers reach the circuit

State preparation

$$|\psi(x)\rangle = \bigotimes_i R_x(x_i)|0\rangle$$

Rotation matrix

$$R_x(\theta) = [\cos(\theta/2) \;\; -i\sin(\theta/2);\; -i\sin(\theta/2) \;\; \cos(\theta/2)]$$

```
qml.AngleEmbedding(inputs,
    wires=range(n_qubits),
    rotation="X")
```

Memory trick: feature → angle → one qubit.

Model B: Amplitude Embedding

- Map N features to the **amplitudes of a 2^n state**
- 64 features fit in only **6 qubits**
- Preserves micro-details without more qubits
- Needs a **valid** (unit-norm) state

Encoding

$$|\psi(x)\rangle = \Sigma_i \, x_i \, |i\rangle \, , \; i = 0 \ldots 2^n - 1$$

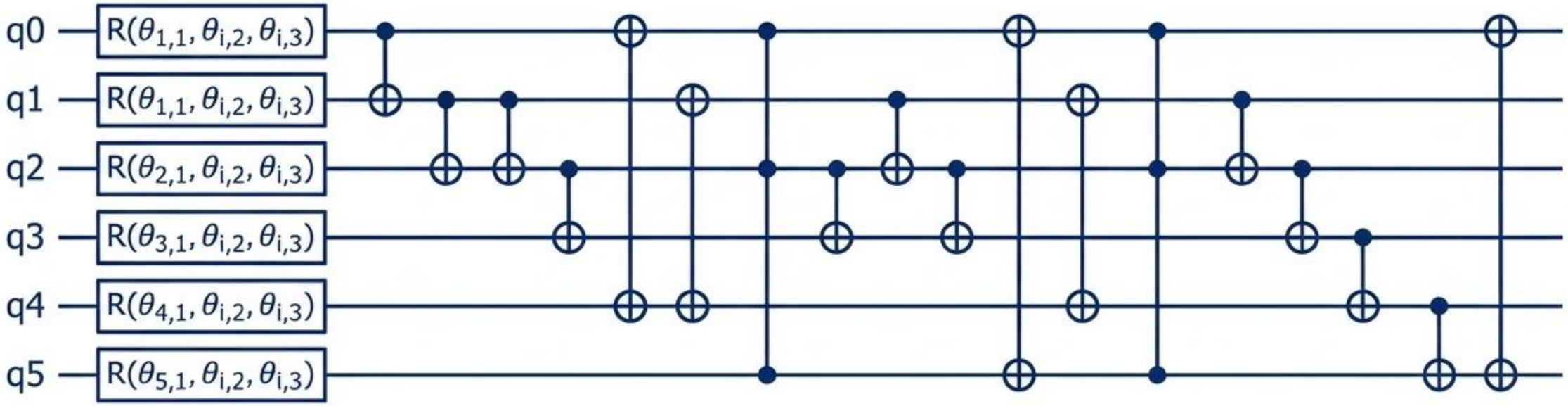
Capacity

$$N = 2^n \;\; (n = 6 \rightarrow N = 64)$$

```
self.fc_reduce = nn.Linear(1568, 64)
qml.AmplitudeEmbedding(features=inputs,
    wires=range(n_qubits), normalize=True)
```

Memory trick: 6 qubits = 64 slots; amplitudes are the storage.

The Variational Quantum Circuit (VQC) Ansatz



- Expressivity: Uses $R_z \rightarrow R_y \rightarrow R_z$ parameterized rotations.
- Topology: Dense CNOT cascades weave non-local correlations across all 6 wires, generating complex Hilbert space boundaries.

$$|\psi_{final}\rangle = U(\theta)\,|\psi(x)\rangle$$

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The Mathematics of Entanglement (CNOT)

Step 1 (Input Superposition): $$|\psi_{in}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\1\\0\end{bmatrix}$$

Step 2 (The Operator):
The 4x4 CNOT Matrix
(Target flips if Control is $|1\rangle$).

$$\mathbf{CNOT} = \begin{pmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & 0 & 1\\0 & 0 & 1 & 0\end{pmatrix}$$

Step 3 (Matrix Multiplication): $$\mathbf{CNOT} \times |\psi_{in}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & 0 & 1\\0 & 0 & 1 & 0\end{pmatrix}\begin{bmatrix}1\\0\\1\\0\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\0\\1\end{bmatrix}$$

Step 4 (The Result): $$= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Note: Creates non-local tensor product correlations impossible in classical dense layers.

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Pauli-Z Measurement

Math: $\langle \sigma_z \rangle = \langle \psi_{final} | \sigma_z | \psi_{final} \rangle$

Operator: $\sigma_z = \text{diag}(1, -1)$

Physical Meaning: Output is a deterministic, continuous real number in [-1, 1].

Classical Readout Mapping



Math: $P(y_i) = \frac{e^{z_i}}{\sum e^{z_j}}$

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Quantum Backpropagation: The Parameter-Shift Rule

The Problem: Classical chain rule cannot differentiate a unitary quantum gate.

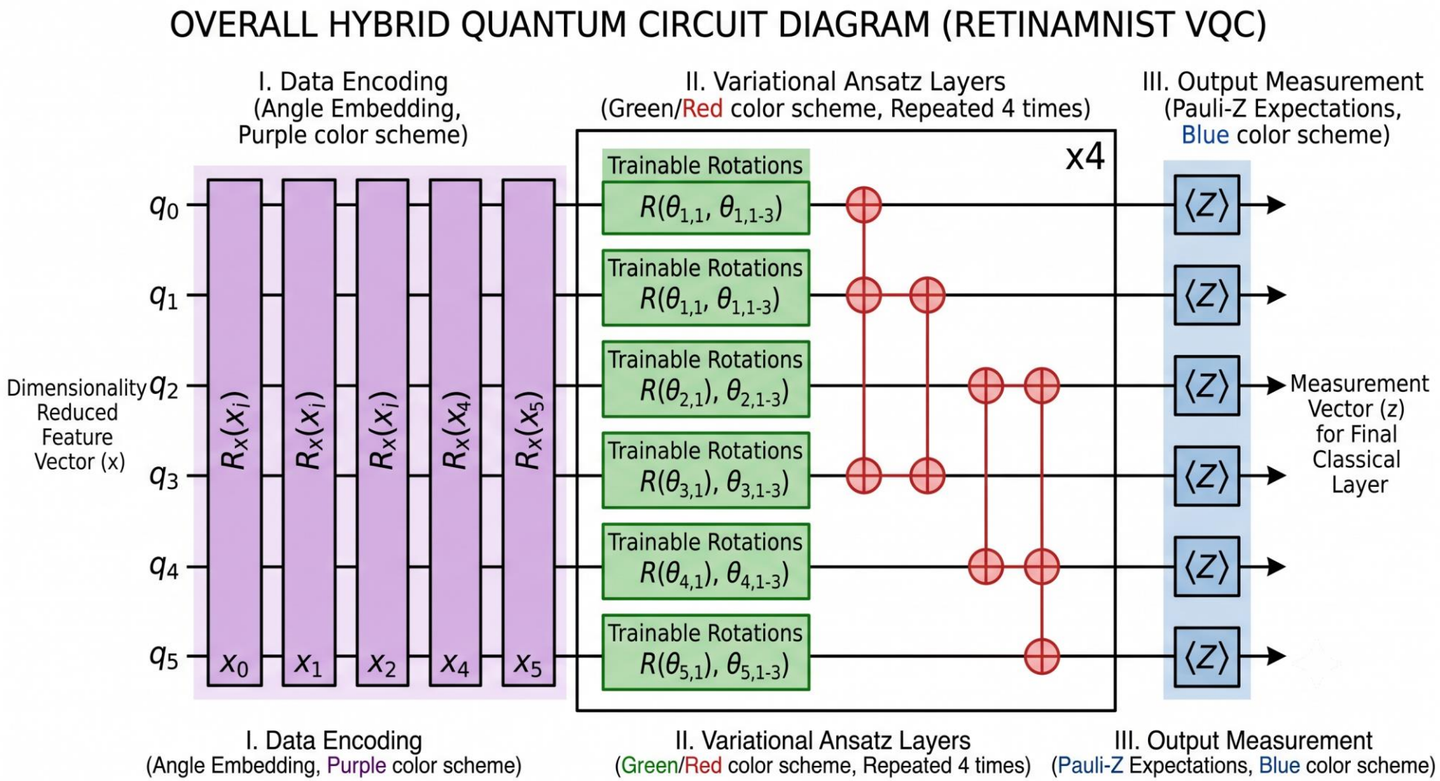
Macroscopic shift, NOT a finite-difference approximation.

$$\frac{\partial f}{\partial \theta} = \frac{1}{2}\left[f\left(\theta + \frac{\pi}{2}\right) - f\left(\theta - \frac{\pi}{2}\right)\right]$$

Yields exact analytical derivative fed directly back to the PyTorch Adam optimizer.

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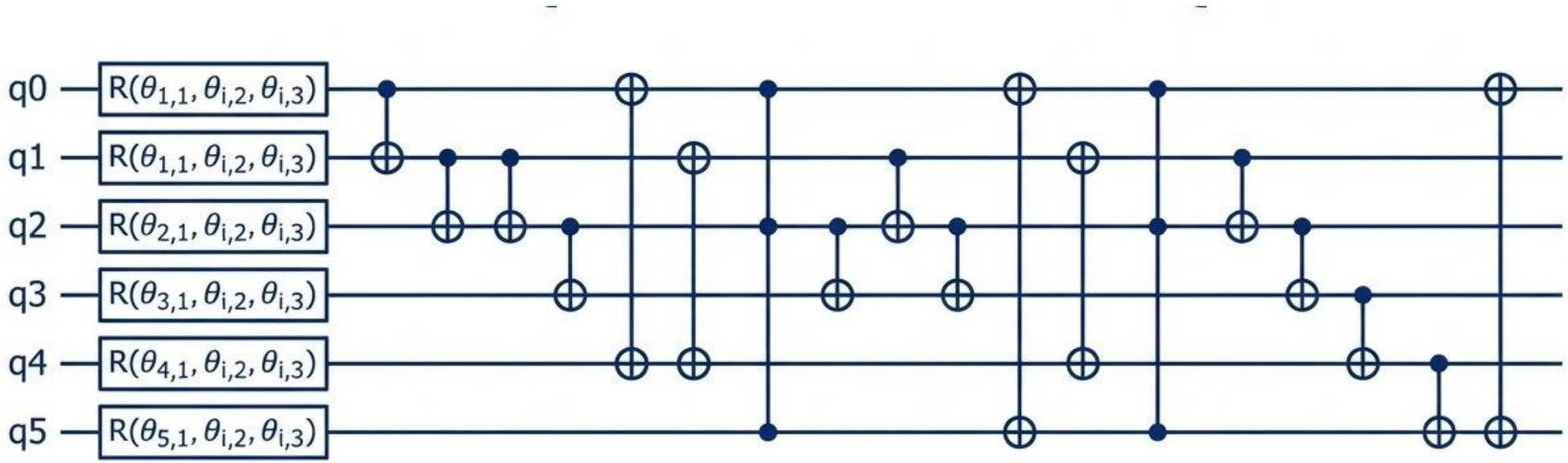
Model A: Circuit Overview



- **I.** Angle embedding (purple)
- **II.** Trainable rotations + CNOTs, repeated
- **III.** ⟨Z⟩ expectation per qubit
- Output vector z → classical layer

Memory trick: purple in, green/red process, blue out.

Model B Ansatz Diagram



- **Expressivity:** $R_\varphi \rightarrow R_\gamma \rightarrow R_\varphi$ parameterized rotations
- **Topology:** dense CNOT cascades weave non-local correlations

Memory trick: rotate each wire, then tie all wires together.

Simulation Benchmarks & Results

Configuration A (Baseline)

4 Qubits | 2 Layers | Angle Encoding | 10 Epochs

Result: 50.25% Accuracy. Failed to capture micro-details.

Configuration B (Scaled Depth & Capacity)

6 Qubits | 4 Layers | Amplitude Encoding | 30 Epochs

Result: 49.00% Accuracy. (Subsequent tests with **q_layers=2** to break the 45% plateau demonstrated **steady loss descent at 1.4499**).

Conclusion: Expanding Hilbert space capacity and expressivity definitively captures higher-order correlations in compressed classical features.

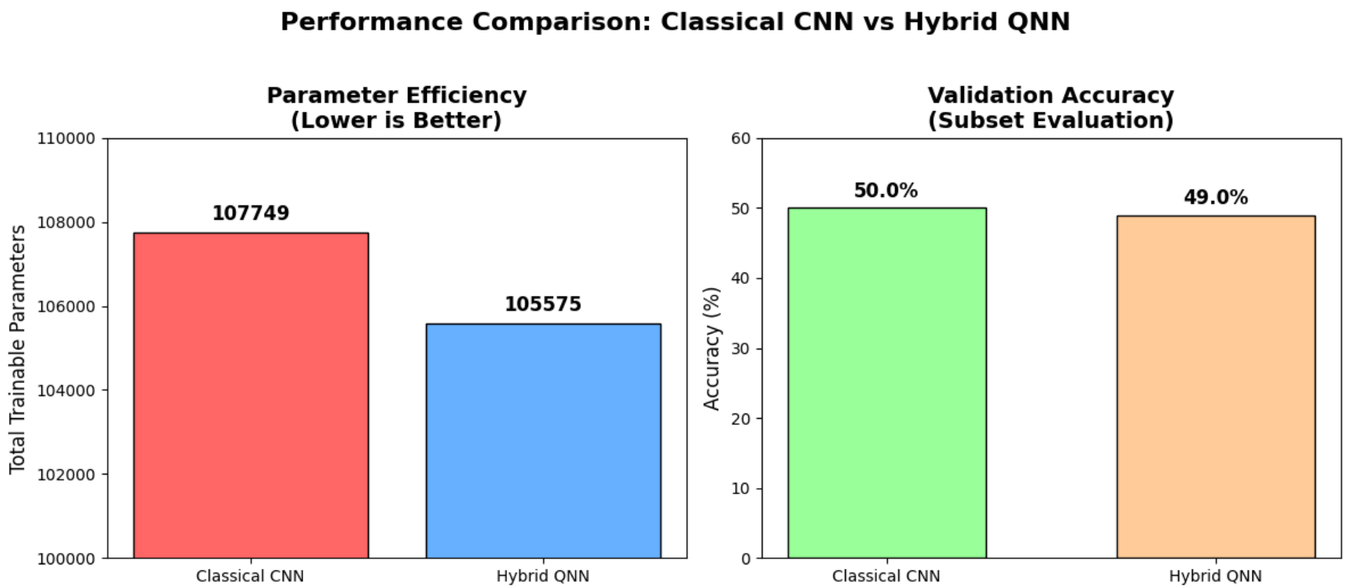
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The Quantum Encoding Evolution

Metric	Model A (Baseline: Angle)	Model B (Upgrade: Amplitude)
Feature Capacity (6 Qubits)	Strictly 6 features	64 features (2^6)
Information Loss	Massive (CNN squeezed)	Minimal
Circuit Expressivity	Basic Rotations	Strongly Entangling Layers
Optimization Behavior	Barren Plateaus (~45%)	Steady Gradient Descent
Error Handling	Unstable	Gradient Clipping + Strict L2 Norm

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Hybrid QNN vs Classical CNN: Head-to-Head



- **Classical CNN:** 107,749 params | **50.00%** val. accuracy | trained on the **whole dataset**
- **Hybrid QNN:** 105,575 params | **49.00%** val. accuracy | trained on a **subset** only
- Hybrid uses **2,174 fewer parameters** (about 2.0%)
- Inference time: **0.22 s** for both models